

Phase 07 Open Boundary: A First-Order ABC for A-Phi

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Phase 07 open boundary: a first-order ABC for A-Phi

Decision

Phase 07's interim, approximate open-boundary treatment is a **first-order absorbing boundary condition (ABC)**, not a PML. This document records the literature check behind that choice and the from-scratch derivation of the ABC written directly in terms of the potentials **A** and **Phi**, since no such formula was found already published.

Why ABC over PML (recap)

PML is the more common default in general full-wave E/H-field codes, but two things push against it here specifically:

1. **PML has known low-frequency degeneracy** (the complex coordinate-stretching parameters become ill-conditioned as $\omega \rightarrow 0$). That would put a second low-frequency failure mode right next to a formulation whose entire premise is all-frequency stability from DC through THz – a direct conflict with the reasoning behind every other choice in this project (keeping the epsilon terms, preferring ACA over MLFMA, etc.).
2. **PML is the larger implementation lift for this specific solver.** It needs an added volumetric absorbing layer (a mesh change), anisotropic complex-tensor assembly inside that layer, and empirical thickness/grading-profile tuning – on top of translating a technique whose published form is for E/H fields onto the coupled A-Phi system while keeping the chosen gauge (Sec. below) valid inside the stretched region. An ABC costs one short field-to-potential substitution and drops in as a single boundary term added to the existing weak form: no new DOFs, no mesh change.

An ABC’s trade-off – accuracy that degrades away from near-normal incidence, and a need for reasonable standoff distance from the scatterer – is a bounded, documentable limitation, not a new failure regime. Phase 12 replaces this whole boundary treatment with an exact FEM-BI coupling once the core solver is proven, so the interim treatment’s job is just to be honest about its error budget, per the roadmap’s own requirement.

Literature check: does a published A-Phi-native ABC exist?

Searched specifically for an absorbing or radiation boundary condition written directly in terms of $\mathbf{A}$ and Φ , rather than applied to derived E/H fields. Result: **no such formula was found**. What exists instead:

- **Zhang, Na, Jiao & Chew, “An A-Phi Formulation Solver in Electromagnetics,” arXiv:2207.02260 (2022)** – the closest existing broadband A-Phi solver (already cited in docs/REFERENCES.md for its DEC discretization, with no boundary-integral coupling). Its own numerical validation does not use an A-Phi-native ABC: a rod-antenna example uses a simple impedance boundary condition (IBC) described as “a simple absorbing boundary condition,” and a nano-scale rod-antenna example uses plain PEC truncation “for simplicity.” PML is discussed narratively but its implementation is deferred to a separate discrete-exterior-calculus reference, with no A-Phi equations given. This is consistent with the gap already noted in docs/FULL_WAVE_SCOPE.md’s novelty note for the BEM-coupling question – the open-boundary treatment for A-Phi solvers generally falls back to simplified field-based conditions or is left unspecified.
- **Ciuprina & Sabriego, “Electric circuit element boundary conditions for electromagneto-quasistatic and full wave models in A, phi potentials,” J. Math. Industry, vol. 14, art. 27 (2024)** – covers port/circuit boundary conditions (equipotential terminals, voltage/current excitation) for A-Phi FEM, not open/radiating boundaries. Not applicable here, but worth knowing about for Phase 06’s port abstraction.
- **Chervyakov, “On the use of mixed potential formulation for finite-element analysis of large-scale magnetization problems,” arXiv:2307.12308 (2023)** – magnetostatic only; truncates with PMC ($\mathbf{n} \times \mathbf{H} = 0$) and magnetic insulation ($\mathbf{n} \cdot \mathbf{B} = 0$) on a large sphere, not a radiation condition. Not applicable to a full-wave radiating problem.

Conclusion: the ABC below is a from-scratch derivation applying a standard, widely-published, peer-reviewed field-based result (the Silver-Muller/Sommerfeld first-order ABC, verified against Chen & Chew 2017, Eq. 83, above) to this project’s own A-Phi system, via the standard potential definitions already in use elsewhere in this roadmap. No employer-authored or otherwise non-public material entered this derivation.

Literature check: PML for A-Phi specifically (Sept 2026)

A separate, dedicated search for a PML formulated on A-Phi potentials (as opposed to E/H fields) turned up nothing published either – if anything, a narrower gap than the ABC case:

- **Chen & Chew, J. Comput. Phys. 350 (2017)** (above) explicitly implements PEC, PMC, Dirichlet, periodic, and ABC boundary treatments in a DEC/E-H setting, but **does not implement PML** – confirmed directly from the paper’s boundary-condition section.
- **Zhang, Na, Jiao & Chew, arXiv:2207.02260 (2022)** discusses PML narratively but defers its actual implementation to reference [9] in that paper’s own bibliography – J. Rabina, “On a numerical solution of the Maxwell equations by discrete exterior calculus,” Ph.D. dissertation, University of Jyväskylä, 2014. That dissertation was not directly accessible to verify, but everything else found about this DEC lineage (Chen & Chew 2017, and Zhang/Na/Jiao/Chew’s own paper) formulates its Maxwell solver in E/H differential forms, not A-Phi – there is no indication anywhere in this citation chain that a

PML was ever written directly on A-Phi potentials, only that a DEC-framework PML exists for the field-based version of the same broader numerical lineage.

- General searches for “vector potential” + PML in the wider literature return eddy-current/magnetostatic vector-potential-only formulations (no coupled scalar potential, no radiation, e.g. A-V or T-Omega eddy-current methods) – a different, non-radiating problem class, not evidence of an A-Phi-plus-PML radiation treatment.

Conclusion: no A-Phi-native PML exists in the literature either. This doesn’t change the Phase 07 decision (ABC over PML) – if anything it reinforces it: adopting PML here would mean adapting an *unpublished-for-A-Phi* technique with a harder physical translation (anisotropic complex-stretched material tensors threaded through the coupled curl-curl/gradient system while preserving the gauge) and a known low-frequency failure mode, versus the ABC’s short, verified field-to-potential substitution.

Why this gap exists (Sept 2026)

Worth recording deliberately, since it bears directly on the Phase 12 “candidate paper” angle already noted in docs/FULL_WAVE_SCOPE.md: the two literature checks above found no A-Phi-native ABC or PML, and there’s a reasoned explanation for that gap – part evidenced, part this project’s own inference, kept clearly separated below.

Evidenced: two communities with non-overlapping needs. Every source found points the same direction. A-Phi’s home has historically been the magnetoquasistatic/eddy-current world (motors, transformers, power electronics, geophysics), where fields don’t radiate by construction, so there was never a reason to develop a radiation boundary for potentials. Meanwhile, every ABC/PML result found stays in E/H fields even when the paper’s whole point is DC-to-high-frequency validity: Chen & Chew (2017) implements ABC on E/H inside a DEC framework; Zhu & Jiao’s “theoretically rigorous full-wave FEM from DC to high frequencies” (IEEE Trans. Adv. Packaging, vol. 33, no. 3, 2010) is *also* E-field, and even it doesn’t implement a true open/radiating boundary – it truncates a packaging geometry with a Neumann-type condition on an air layer, because its target problems are closed packaging structures, not radiating ones. So the DC-to-high-frequency crowd and the radiating-antenna crowd are two different literatures, and A-Phi historically sits with neither: it wasn’t built by the packaging crowd (who stayed in E-field and didn’t need a radiation boundary anyway) or the antenna crowd (who never needed DC stability). Ansys’s own product split – Maxwell (A-Phi, magnetoquasistatic, no radiation) versus HFSS (E-field, full-wave, proper radiation boundaries) – is the commercial expression of the same non-overlap: nobody who had A-Phi also needed radiation, and nobody who needed radiation had a reason to touch potentials.

This project’s own inference, not sourced to a citation: two secondary factors plausibly compound the gap, offered here as reasoning rather than a literature finding.

1. *Gauge non-uniqueness makes a “generic” A-Phi boundary condition less publishable than the E-field version.* E is physical and gauge-invariant, so one Silver-Muller formula covers every case. A alone is not physical – it only means something once a gauge is fixed. The condition derived above assumes no gauge (it uses only the potential definitions), but a researcher wanting a single, general-purpose closed-form result would need to either commit to a specific gauge or carry it as a parameter, which is a less clean, less citable contribution than the field-based version.
2. *The derivation is arguably too mechanical, on its own, to be paper-worthy.* What this document does – substitute known potential definitions into an already-published equation – is a page of algebra applied to existing results, not new physics. Anyone who needed this before likely did the same substitution privately (as this project just did) without publishing it separately, since it doesn’t read as a novel enough contribution on its own to submit anywhere.

Ties back to this project's own novelty case. The same underlying cause already identified in docs/REFERENCES.md's novelty note for the BEM-coupling question applies here too: nobody needed an all-frequency-stable A-Phi system with a real open/radiating boundary until a problem class required both ends of the spectrum in the *same* solve. Sub-micron THz phased-array feeds co-designed with electrically large radiating apertures is exactly that kind of problem. The ABC gap and the BEM-coupling gap are plausibly the same gap, found twice, from two different angles – worth keeping in mind together when scoping the Phase 12 standalone-paper angle, rather than treating them as two unrelated findings.

Derivation

Starting point: the standard field-based first-order ABC

The standard first-order (Silver-Muller-type) absorbing boundary condition, applied on a truncation surface Γ_{∞} enclosing all sources/scatterers, is

$$\mathbf{n} \times (\text{curl } \mathbf{E}) - j k [\mathbf{n} \times (\mathbf{n} \times \mathbf{E})] = 0 \quad \text{on } \Gamma_{\infty}$$

with $\mathbf{n}$ the outward unit normal, $k = \omega \sqrt{\mu \epsilon}$ the free-space wavenumber, and the $e^{+j\omega t}$ time convention used throughout this project's equations (matching docs/ROADMAP.md and docs/FULL_WAVE_SCOPE.md). This condition is exact for an outgoing spherical wave in the far field and approximate at finite radius; its accuracy degrades for non-normal incidence, which is the ABC's known, bounded limitation referenced above.

Verified against a specific peer-reviewed source (Sept 2026 update): this exact 3-D first-order condition, $\mathbf{n} \times (\text{curl } \mathbf{E}) = j\omega\mu(\mathbf{n} \times \mathbf{H}) \sim -jk[\mathbf{n} \times (\mathbf{n} \times \mathbf{E})]$, appears as Eq. (83) in Chen & Chew, "Numerical Electromagnetic Frequency Domain Analysis with Discrete Exterior Calculus," J. Comput. Phys., vol. 350, pp. 668-689, 2017 (arXiv:1704.05145) – quoted there in the opposite (physics, $e^{-i\omega t}$) time convention, which is why their sign on the k term is $+i$ where this document's $e^{+j\omega t}$ convention gives $-j$; the physical content is identical. That paper (by W. C. Chew, the same author behind the generalized Lorenz gauge chosen in Phase 03) implements exactly this ABC in a differential-forms/DEC setting, but formulated on $\mathbf{E}$ and $\mathbf{H}$, not on $\mathbf{A}$ and Φ – confirming the gap this document's derivation fills. It explicitly does not implement PML; see the PML literature check below.

Substituting the potential definitions

This project already defines the fields in terms of the potentials as

$$\mathbf{E} = -j\omega\mathbf{A} - \text{grad}(\Phi)$$

$$\mathbf{B} = \text{curl } \mathbf{A} \quad \Rightarrow \quad \text{curl } \mathbf{E} = \text{curl}(-j\omega\mathbf{A} - \text{grad}(\Phi)) = -j\omega(\text{curl } \mathbf{A})$$

(the $\text{curl}(\text{grad } \Phi) = 0$ identity removes Φ from $\text{curl } \mathbf{E}$ directly, which is exactly why a field-based ABC's Φ -dependence has been hiding inside $\mathbf{E}$ all along, not because Φ has no boundary role here.)

Substitute both into the field-based ABC:

$$\mathbf{n} \times [-j\omega(\text{curl } \mathbf{A})] - jk[\mathbf{n} \times (\mathbf{n} \times (-j\omega\mathbf{A} - \text{grad}(\Phi)))] = 0$$

Expand the second bracket:

$$\mathbf{n} \times (\mathbf{n} \times (-j\omega\mathbf{A} - \text{grad}(\Phi))) = -j\omega[\mathbf{n} \times (\mathbf{n} \times \mathbf{A})] - [\mathbf{n} \times (\mathbf{n} \times \text{grad}(\Phi))]$$

so the equation becomes

$$-j\omega[\mathbf{n} \times (\text{curl } \mathbf{A})] - jk\{-j\omega[\mathbf{n} \times (\mathbf{n} \times \mathbf{A})] - [\mathbf{n} \times (\mathbf{n} \times \text{grad}(\Phi))]\} = 0$$

$$-j*\omega*[n \times (\text{curl } A)] - k*\omega*[n \times (n \times A)] + j*k*[n \times (n \times \text{grad}(\Phi))] = 0$$

Divide through by $-j*\omega$ (nonzero for $\omega \neq 0$; see the DC caveat below):

$$n \times (\text{curl } A) - j*k*[n \times (n \times A)] - (k/\omega)*[n \times (n \times \text{grad}(\Phi))] = 0$$

Result

$$\begin{array}{l} +-----+ \\ | \quad n \times (\text{curl } A) - j*k*[n \times (n \times A)] - (k/\omega)*[n \times (n \times \text{grad } \Phi)] = 0 \quad | \\ | \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad | \\ | \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad | \\ | \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad | \\ +-----+ \end{array}$$

This is a genuinely coupled boundary condition: even though the field-based ABC has no explicit Φ term (it was absorbed into E), decomposing back into the two potential unknowns produces a condition that mixes the tangential curl of A , the tangential component of A itself, and the tangential component of $\text{grad}(\Phi)$. All three terms need to be assembled at the truncation boundary in Phase 07 – dropping the $\text{grad}(\Phi)$ term would silently reduce this back to an incomplete condition.

Weak-form boundary term

For assembly, this enters the weak form the same way any Robin-type natural boundary condition does: test against a vector test function w in the A -space and integrate over Γ_{inf} ,

$$\begin{aligned} \text{integral}_{\{\Gamma_{\text{inf}}\}} w \cdot [n \times (\text{curl } A)] \, dS \\ = \text{integral}_{\{\Gamma_{\text{inf}}\}} w \cdot \{ j*k*[n \times (n \times A)] + (k/\omega)*[n \times (n \times \text{grad } \Phi)] \} \, dS \end{aligned}$$

with the left-hand side being exactly the natural boundary term that already appears from integrating the curl-curl term by parts in Phase 04's weak form – so this ABC replaces the zero-Neumann (or PEC) boundary term with this Robin-type expression, rather than requiring a new integration-by-parts step.

Refined assembly: a symmetric surface-gradient coupling block (Sept 2026 update)

An independent review of this derivation (requested from Google Gemini, then checked against this document before being accepted – same practice already used for the earlier Gemini review noted in docs/REFERENCES.md's terahertz phased-array section) identified a cleaner way to assemble the third, Φ -coupling term above than leaving it as a raw $\text{grad } \Phi$ surface integral.

The tangential double-cross-product of a surface gradient reduces to the negative surface (tangential) gradient itself:

$$n \times (n \times \text{grad } \Phi) = -\text{grad}_{\Gamma}(\Phi)$$

where grad_{Γ} denotes the gradient restricted to the tangent plane of Γ_{inf} . Substituting this into the right-hand side's third term and integrating that piece by parts over the closed surface Γ_{inf} (which has no boundary of its own, so no extra edge term appears) moves the derivative off Φ and onto the test function:

$$\begin{aligned} \text{integral}_{\{\Gamma_{\text{inf}}\}} w \cdot [-(k/\omega) * \text{grad}_{\Gamma}(\Phi)] \, dS \\ = (k/\omega) * \text{integral}_{\{\Gamma_{\text{inf}}\}} [\text{div}_{\Gamma}(w)] * \Phi \, dS \end{aligned}$$

with div_{Γ} the surface divergence of the tangential test function w .

This is a genuine implementation improvement, not just an algebraic rearrangement: it turns the Φ -coupling term into a symmetric bilinear block between the discrete surface divergence of the Whitney (curl-conforming) edge elements already used for $\mathbf{A}$ and the nodal (H1) basis functions already used for Φ , evaluated with the *same* Γ_{inf} surface mesh and the *same* quadrature rule used for the rest of the boundary term – rather than assembling a raw tangential-gradient-of- Φ term against a vector test function, which needs care to keep consistent with how $\text{grad}(\Phi)$ is discretized elsewhere in the volume. Because both sides reduce to the same kind of edge-to-node coupling integral, this keeps the boundary matrix exactly consistent with the interior variational formulation and its existing edge/nodal basis pairing, with no new element type or extra consistency check required at the truncation surface.

The full boxed result from the earlier section is unchanged by this – this is a weak-form assembly refinement of the third term only, applied after testing and integrating, not a change to the strong-form boundary condition itself.

DC caveat

Dividing by $-j \cdot \omega$ above means this condition is undefined at $\omega = 0$, exactly like Chew’s generalized Lorenz gauge chosen for Phase 03 (see docs/REFERENCES.md, “Gauge for mixed conductor/dielectric ports”). This is expected and consistent with the rest of the project: Phase 01, step 3 already commits to a dedicated magnetostatic solve at $\omega = 0$ rather than a naive limit of the frequency-domain system, and an open-boundary radiation condition is not physically meaningful at DC anyway (there is no radiating wave to absorb).

Implementation note

Document, per the roadmap’s own requirement, the standoff distance from the scatterer/source region to Γ_{inf} used in the Phase 07 test problem, and report the observed error against the analytical solution as a function of incidence angle – this is what turns “an ABC has angle-dependent accuracy” from a caveat into an actual, citable number for this project’s own benchmark suite (Phase 08).

References

- S. Chen, W. C. Chew, “Numerical Electromagnetic Frequency Domain Analysis with Discrete Exterior Calculus,” J. Comput. Phys., vol. 350, pp. 668-689, 2017 (arXiv:1704.05145) – Eq. (83) is the specific, verified 3-D first-order Silver-Muller ABC (on E, H) this derivation starts from. Implements PEC/PMC/Dirichlet/periodic/ABC; explicitly does not implement PML.
- A. F. Peterson, S. L. Ray, R. Mittra, *Computational Methods for Electromagnetics*, IEEE Press – general Silver-Muller/Sommerfeld ABC background (already cited in docs/REFERENCES.md for EFIE/MFIE/CFIE background).
- J. Rabina, “On a numerical solution of the Maxwell equations by discrete exterior calculus,” Ph.D. dissertation, University of Jyväskylä, 2014 – the PML-in-DEC reference that Zhang, Na, Jiao & Chew (2022) defer to; not directly verified (dissertation not accessible), and in any case part of the same E/H-based DEC lineage, not A- Φ .
- B. Zhang, D.-Y. Na, D. Jiao, W. C. Chew, “An A- Φ Formulation Solver in Electromagnetics,” arXiv:2207.02260, 2022 – literature-check reference; uses IBC/PEC truncation, not an A- Φ -native ABC.
- G. Ciuprina, R. V. Sabriego, “Electric circuit element boundary conditions for electromagneto-quasistatic and full wave models in A, Φ potentials and their finite element implementation,” J. Math. Industry, vol. 14, art. 27, 2024 – literature-check reference; port/circuit BCs only.

- A. Chervyakov, “On the use of mixed potential formulation for finite-element analysis of large-scale magnetization problems with large memory demand,” arXiv:2307.12308, 2023 – literature-check reference; magnetostatic PMC/insulation truncation only.
- The $E = -dA/dt - \text{grad}(\Phi)$ and $B = \text{curl } A$ potential definitions used above are the same ones already in use throughout this project’s formulation (see docs/ROADMAP.md, Phase 01).
- J. Zhu, D. Jiao, “A Theoretically Rigorous Full-Wave Finite-Element-Based Solution of Maxwell’s Equations from dc to High Frequencies,” IEEE Trans. Adv. Packaging, vol. 33, no. 3, 2010 – “why this gap exists” reference; E-field (not A-Phi), DC-to-high-frequency, and truncates a packaging geometry with a Neumann-type condition rather than a true radiation boundary – evidence that even the DC-to-high-frequency literature has stayed in closed (non-radiating) problem classes.
- Independent review conducted with Google Gemini, Gemini/Open_Boundary_ABC_vs_PML_Evaluation.tex (and compiled PDF) – a secondary synthesis of this document’s own derivation and the public literature above, not an independent source; checked against this document’s derivation (Sept 2026) before folding in the surface-gradient weak-form refinement above, per this project’s standard practice for Gemini-sourced reviews (see docs/REFERENCES.md, terahertz phased array scope note).